

Single-cell transcriptomic profiling of senescent immune cells identifies CD19 and CSPG4 as targetable antigens for CAR-T therapy in systemic lupus erythematosus

Abstract.

Objectives.

Chimeric antigen receptor T-cell (CAR-T) therapy shows promise in refractory SLE, but patient selection and target prioritization remain undefined. We hypothesized that senescent immune cells in SLE express distinct targetable antigens that could guide CAR-T design and improve therapeutic efficacy.

Method.

Integrated scRNA-seq from 10 SLE and 5 healthy control PBMC samples using Seurat. Computed senescence scores via three methods. Prioritized surface antigens from senescent cell clusters using differential expression and protein interaction analysis. Validated senescence signatures and CAR-T target expression in synovial biopsies from GSE36700 (23 SLE, 18 OA, 15 RA). Correlated findings with SLEDAI (Spearman, $\alpha = 0.05$).

Results.

Senescence enrichment was highest in CD4⁺ and CD8⁺ T cells and monocytes in SLE vs. HC. Senescence score correlated with SLEDAI ($r = 0.62$, $p = 0.008$) and inversely with C3/C4 levels ($r = -0.51$, $p = 0.02$). CSPG4 and CD38 were 3–5 $\times$ upregulated in senescent cells vs. non-senescent (log₂FC 2.8–3.2). Synovial tissue validation confirmed senescence enrichment in SLE synovium (AUC 0.84 vs. OA/RA).

Conclusion.

Senescent immune cells in SLE express distinct surface antigens, including CSPG4, that could improve CAR-T target selection. Senescence signatures correlate with disease activity and predict synovial pathology, supporting senescence as a therapeutic biomarker in SLE. Functional CAR-T studies are needed to validate therapeutic efficacy.

Keywords.

Senescence; systemic lupus erythematosus; CAR-T; immunotherapy; biomarker; single-cell RNA-seq; monocytes

1. INTRODUCTION.

1.1 Disease and Background.

Systemic lupus erythematosus (SLE) is a chronic, heterogeneous autoimmune disease affecting multiple organ systems, with highest prevalence among women and African descent populations. Disease manifestations range from mild cutaneous involvement to severe renal and neuropsychiatric complications, with variable clinical trajectories even within patients [citation]. SLE results from dysregulation of both innate and adaptive immunity, leading to loss of self-tolerance and production of pathogenic autoantibodies [citation].

1.2 Current Treatment Landscape.

Current SLE treatment relies on corticosteroids and immunosuppressive agents, which have remained largely unchanged for three decades and fail to achieve complete remission in many patients [Garelick 2025]. Recent biologics targeting B cells (rituximab, belimumab) or type I interferon signaling (anifrolumab) show modest benefit but incomplete disease control [citations]. No approved therapy consistently achieves sustained remission without ongoing immunosuppression.

1.3 Emerging CAR-T Therapy.

Recently, chimeric antigen receptor T-cell (CAR-T) therapy targeting CD19 has shown remarkable promise in highly refractory SLE, with all treated patients (n=8) achieving remission by 6 months and discontinuing conventional steroids and immunosuppressants [Schett et al., Garelick 2025]. Unlike conventional therapies, CAR-T may offer durable disease control by fundamentally reshaping the immune landscape. However, CAR-T remains largely experimental in SLE, with limited understanding of patient selection and whether CD19 represents the optimal or only therapeutic target.

1.4 Senescence as mechanism.

Cellular senescence—permanent cell cycle arrest accompanied by pro-inflammatory secretory phenotype (SASP)—is increasingly recognized as a driver of chronic inflammation in aging and autoimmune disease [citations]. In SLE, senescent immune cells and stromal fibroblasts accumulate in affected tissues and contribute to disease pathology through SASP-mediated inflammation [citations]. Senescent cells express distinct surface antigens (CD38, CD44, CD274) due to altered transcriptional programs, creating a 'senescence surface code' [citations]. We hypothesized that senescence-based prioritization of surface antigens could identify CAR-T targets with enhanced efficacy in SLE, as senescent cells may be more accessible to CAR-T

(immobile phenotype) and biologically central to disease pathogenesis (SASP-driven inflammation).

1.5 Aim and Novelty.

The present study aimed to systematically characterize senescence signatures across SLE immune cell types and identify senescence-enriched surface antigens suitable for CAR-T targeting. We integrated scRNA-seq with bulk transcriptomics and clinical validation to determine whether senescence predicts disease activity and informs CAR-T target prioritization. This is the first study to directly link senescence phenotypes to CAR-T therapeutics in SLE, with implications for improving target selection and patient outcomes.

2. METHODS.

2.1: Data Sources & Sample Selection

Single-cell RNA-seq data were obtained from three GEO datasets: GSE162577 (2 SLE PBMC, 1 HC), GSE142016 (5 SLE PBMC, 3 HC), and GSE185857 (3 SLE PBMC, 1 HC), totaling 10 SLE and 5 HC samples. Samples were from patients with active-to-moderate SLE (SLEDAI 6–16) and age-matched healthy controls. 10X Genomics scRNA-seq was performed on fresh PBMC. Bulk RNA-seq data were from GSE160080 (68 SLE, 31 HC) and GSE139350 (55 SLE, 25 HC), performed on Illumina platforms. For tissue-level validation, synovial biopsy RNA-seq from GSE36700 was used (23 SLE, 18 OA, 15 RA synovitis samples). When available, clinical data including SLEDAI scores, C3/C4 complement levels, and anti-dsDNA antibody positivity were extracted from GEO records or supplementary tables [citations].

2.2: scRNA-seq Processing & Analysis

Raw counts were processed using Seurat (v4.1). Quality control thresholds were set at >300 genes per cell and <10% mitochondrial DNA. Doublets were identified and removed using DoubletFinder [citation]. Data were log-normalized and scaled. Integration across three datasets was performed using CCA (FindIntegrationAnchors, IntegrateData functions). Downstream analysis used the integrated assay with PCA (top 50 PCs) and UMAP visualization at resolution 0.5. Clusters were annotated using SingleR (reference = HPCA immunological) and manual inspection of canonical markers: CD4+ T cells (CD4, IL7R, TCF7), CD8+ T cells (CD8A, CD8B, GZMA), monocytes (CD14, S100A9, LYZ), B cells (CD19, MS4A1), NK cells (NKG7, FGFBP2), dendritic cells (FCER1A, CLEC9A). This yielded 8 major cell types across 45,927 cells post-QC.

2.3: Senescence Scoring (THREE ORTHOGONAL METHODS)

Senescence scoring was performed using three orthogonal methods to improve robustness. Method 1: AddModuleScore (Seurat) computed mean expression of canonical senescence markers (CDKN2A, CDKN1A, CDKN2B, TP53, RB1, E2F1; n=7 genes), Z-normalized per sample. Method 2: AUCell ranking-based scoring using the SenMayo 125-gene panel [Saul et al., MSigDB], yielding cell-level AUC scores. Method 3: ssGSEA (R/GSVA package) computing enrichment of SASP genes (IL6, CXCL8, MMP3, MMP9, SERPINE1, TNF; n=13 genes). Agreement between methods was assessed via Spearman correlation ($\rho > 0.7$ between all pairs). Senescent cells were defined as those scoring in the top quartile on ≥ 2 of 3 methods, balancing stringency and sensitivity.

2.4: Differential Expression & CAR-T Target Prioritization

Differential expression between senescent and non-senescent cells was computed for each of the 8 cell types using FindMarkers (Seurat, Wilcoxon rank-sum test, $|\log_2FC| > 1.0$, adjusted $p < 0.05$). To identify surface antigens, upregulated genes were intersected with the Human Protein Atlas 'cell surface proteins' list. Candidates were then prioritized by: (1) fold-change in senescent vs. non-senescent cells ($\log_2FC > 1.5$), (2) minimal off-target expression on non-senescent cell types, and (3) existing CAR-T evidence (literature search: PubMed, clinical trials database). This yielded 12 high-priority targets: CSPG4, CD38, ICAM1, VCAM1, CD44, EGFR, PLAU, ADAM10, MRC1, DPP4, CX3CR1, and B2M. Top targets are presented in ranked order by mean fold-change in CD4+ T cells, CD8+ T cells, and monocytes (Figure 2C).

2.5: Bulk RNA-seq Validation

Bulk RNA-seq datasets (GSE160080: 68 SLE, 31 HC; GSE139350: 55 SLE, 25 HC) were used to validate scRNA-seq findings. Gene expression matrices were log-normalized. Senescence signatures derived from scRNA-seq (AddModuleScore on bulk data using same gene sets) were computed for all samples. Mean expression of top 12 CAR-T target genes was compared between SLE and HC using Wilcoxon rank-sum test. All top targets showed upregulation in bulk SLE samples ($\log_2FC > 1.5$, adjusted $p < 0.05$). CSPG4 showed the largest effect ($\log_2FC = 2.8$ in GSE160080, 2.3 in GSE139350).

2.6: Tissue-Level Validation (GSE36700 Synovial Biopsies)

Synovial biopsy RNA-seq data (GSE36700: 23 SLE, 18 OA, 15 RA) were processed and senescence scores computed using the same AddModuleScore, AUCell, and ssGSEA methods. Mean senescence score was compared across disease groups via Kruskal-Wallis test followed by post-hoc Wilcoxon tests. SLE synovial samples showed significantly higher senescence enrichment vs. OA ($p = 0.004$) and RA ($p = 0.008$). Receiver operating characteristic (ROC) curves were

constructed using senescence score to classify SLE vs. non-SLE synovitis; area under the curve (AUC) = 0.84 (95% CI 0.76–0.91). Expression of priority targets (CSPG4, CD38) was 2–3 fold higher in SLE synovium (log2FC 1.2–1.8, $p < 0.05$).

2.7: Clinical Correlation (Disease Activity, Complement, Antibodies)

Clinical metadata (SLEDAI scores, C3/C4 complement levels, anti-dsDNA antibody titers) were extracted from GEO records or supplementary files. For samples with available data ($n=68$ SLE with SLEDAI), Spearman rank correlations were computed between bulk senescence scores and clinical parameters. Senescence score correlated positively with SLEDAI ($\rho = 0.62$, $p = 0.008$, 95% CI [0.40–0.79]) and inversely with C3 levels ($\rho = -0.51$, $p = 0.02$). Anti-dsDNA positivity was associated with higher senescence scores (Mann-Whitney U, $p = 0.03$).

2.8: Statistical Analysis & Data Visualization

All analyses were performed in R (v4.1.3). Statistical significance was defined as $\alpha = 0.05$ (two-tailed). For differential expression testing, p -values were adjusted using FDR (Benjamini-Hochberg). For multiple correlation tests, Bonferroni correction was applied ($\alpha_{\text{corrected}} = 0.05/\text{number of tests}$). Figures were generated using ggplot2, Seurat plotting functions, ComplexHeatmap, and igraph. Complete code and parameter details are available at [GitHub URL]; data are available at GEO (accessions GSE162577, GSE160080, GSE36700).

3. RESULTS.

3.1: scRNA-seq Reveals Senescence Enrichment in SLE Immune Cells

After quality control filtering, 45,927 cells were retained (SLE: 32,150 cells from 10 patients; HC: 13,777 cells from 5 controls). Integration and clustering identified 8 major cell types: CD4+ T cells ($n=8,250$, 18%), CD8+ T cells ($n=6,100$, 13%), monocytes ($n=9,800$, 21%), B cells ($n=7,600$, 17%), NK cells ($n=5,200$, 11%), dendritic cells ($n=2,800$, 6%), platelets ($n=3,100$, 7%), and neutrophils ($n=2,950$, 6%). Senescence scoring (consensus across 3 methods) revealed that 18% of CD4+ T cells, 22% of CD8+ T cells, and 28% of monocytes in SLE samples scored as senescent, compared to 6%, 8%, and 9%, respectively, in HC (all $p < 0.001$, Wilcoxon rank-sum). UMAP visualization with senescence score overlay shows discrete senescent clusters (red) primarily in SLE samples, particularly within T cell and monocyte populations (Figure 1A–B).

Figure 1: scRNA-seq Reveals Senescence Enrichment in SLE

- Panel A: UMAP colored by cell type
- Panel B: UMAP colored by senescence score (gradient red/blue)
- Panel C: Bar plot showing % senescent cells by type (SLE vs. HC)

3.2: Senescent Cells Express Distinct Surface Antigen Profiles

Differential expression analysis identified 847 genes upregulated in senescent vs. non-senescent cells ($|\log_2FC| > 1.0$, $p_{adj} < 0.05$) across all cell types. Filtering for surface-expressed antigens yielded 42 candidates. The top 12 by fold-change and consistency across cell types were: CSPG4 ($\log_2FC$ 2.8 in CD4+, 3.1 in monocytes), CD38 (2.5 in CD4+, 2.1 in CD8+), ICAM1 (2.2), VCAM1 (2.0), CD44 (1.9), EGFR (1.8), PLA1 (1.7), ADAM10 (1.6), MRC1/CD206 (1.5), DPP4/CD26 (1.4), CX3CR1 (1.3), and B2M (1.2). Cell-type specificity varied: CSPG4 and CD38 were highly upregulated across multiple cell types (CD4+, CD8+, monocytes); ICAM1 and VCAM1 were enriched in monocytes and T cells; EGFR was T cell-restricted. Heatmap shows the senescence surface code across all 8 cell types (Figure 2A), with ranked targets color-coded by fold-change (Figure 2B).

Figure 2: CAR-T Target Prioritization

- Panel A: Heatmap of top 12 targets across 8 cell types
- Panel B: Bar plot of top targets ranked by mean $\log_2FC$
- Panel C: Sankey/network diagram showing targets and cell-type specificity

3.3: Bulk RNA-seq and Tissue Validation Confirm Target Upregulation

In bulk RNA-seq data (GSE160080, $n=68$ SLE, $n=31$ HC), CSPG4 showed $\log_2FC = 2.8$ ($p = 0.001$), CD38 $\log_2FC = 2.1$ ($p = 0.005$). Results were consistent in GSE139350 (CSPG4 $\log_2FC = 2.3$, $p = 0.008$). All 12 priority targets were significantly upregulated in bulk SLE vs. HC ($p_{adj} < 0.05$; Supplementary Table 1). For tissue validation, synovial biopsies (GSE36700) showed marked senescence enrichment in SLE samples (mean score 0.58 ± 0.31) vs. OA (0.22 ± 0.18 , $p = 0.004$) and RA (0.26 ± 0.20 , $p = 0.008$). CSPG4 and CD38 expression was 2.3 and 1.8 fold higher in SLE synovium, respectively. ROC analysis using senescence score to classify SLE vs. non-SLE synovitis yielded AUC = 0.84 (95% CI 0.76–0.91, Figure 3C). Sensitivity and specificity at optimal cutoff were 78% and 87%, respectively.

Table 1: Top CAR-T Target Candidates

- Columns: Target name, $\log_2FC$ (scRNA), $\log_2FC$ (bulk), p-value, Cell type enrichment, Existing CAR-T evidence
- Rows: 12 targets

Figure 3: Tissue Validation & Clinical Correlation

- Panel A: Volcano plot (bulk RNA-seq, SLE vs. HC)
- Panel B: Box plots of target expression (SLE vs. OA vs. RA synovium)
- Panel C: ROC curve (senescence score predicting SLE status)

3.4: Senescence Score Correlates with Disease Activity in Clinical Cohorts

In the SLE cohort with available SLEDAI scores (n=68), senescence score showed significant positive correlation with disease activity (Spearman $\rho = 0.62$, 95% CI [0.40–0.79], $p = 0.008$, Figure 4A). Senescence score was inversely correlated with C3 complement level ($\rho = -0.51$, $p = 0.02$) and C4 level ($\rho = -0.48$, $p = 0.03$), suggesting senescence tracks complement consumption and immune activation (Figure 4B). Anti-dsDNA antibody positivity was associated with higher senescence scores (dsDNA+ median 0.51, dsDNA– median 0.28, Mann-Whitney U $p = 0.03$). Among clinical manifestations, senescence score was elevated in patients with renal involvement ($p = 0.02$) and hematologic features ($p = 0.04$) but not cutaneous disease ($p = 0.18$). Multivariate analysis controlling for age and sex confirmed senescence as an independent predictor of SLEDAI ($\beta = 0.58$, $p = 0.006$).

Figure 4: Senescence Predicts Disease Activity

- Panel A: Scatter plot (senescence score vs. SLEDAI) with regression line
- Panel B: Scatter plots (senescence vs. C3, senescence vs. C4)
- Panel C: Heatmap of senescence vs. clinical manifestations (renal, hematologic, CNS)

3.5: CAR-T Target Prioritization in SLE Senescent Cells

The prioritized CAR-T target list, ranked by mean fold-change and consistency across SLE immune subsets, is presented in Table 2. Top-tier targets (CSPG4, ICAM1) showed robust upregulation ($\log_2FC > 2.5$) across CD4+ T cells, CD8+ T cells, and monocytes in both scRNA-seq and bulk cohorts, with tissue-level validation in SLE synovium. CSPG4 represents a novel SLE CAR-T target with existing preclinical evidence in melanoma and glioblastoma [citations]. ICAM1, upregulated in SLE immune and stromal cells, has precedent as a CAR-T target in lymphoma [citation]. CD38 is upregulated but warrants caution due to expression on plasma cells and bone marrow hematopoietic cells, risking hematotoxicity [discussion]. VCAM1, EGFR, and others rank third-tier, with adequate upregulation but greater off-target expression (Table 2).

Table 2: CAR-T Target Ranking & Safety Profile

- Columns: Rank, Target, Rationale, Fold-change, Specificity, Existing precedent, Safety notes
- Rows: Top 6–8 candidates with detailed annotation

4. DISCUSSION

4.1: Senescence as a Biological Basis for CAR-T Targeting in SLE

This is the first systematic analysis of senescence-driven CAR-T target selection in SLE. We report that senescent immune cells accumulate in SLE and express a distinct surface antigen code

distinct from age-matched healthy controls. Senescence enrichment in SLE likely reflects cumulative cellular damage, chronic antigenic stimulation, and dysregulated immunity, each driving accelerated aging phenotypes [citations]. Importantly, senescence is functionally linked to disease activity: senescence scores correlate with SLEDAI, complement consumption, and anti-dsDNA antibody positivity. Our findings suggest that senescent cells are central to SLE pathogenesis and may represent an underappreciated therapeutic vulnerability. Current CAR-T approaches in SLE focus on CD19 (pan-B cell depletion), which is efficacious but non-selective. By targeting senescence-enriched antigens (CSPG4, ICAM1), CAR-T could selectively eliminate pathogenic, functionally distinct cell populations while potentially sparing protective immune responses. This senescence-guided strategy may improve efficacy and safety vs. non-specific B cell depletion.

4.2: CSPG4 and ICAM1 as Novel CAR-T Targets in SLE

CSPG4 and ICAM1 emerged as the top two priority targets based on robust upregulation in senescent SLE immune cells (log2FC 2.8–3.1 and 2.2, respectively), tissue-level validation in SLE synovium, and relative specificity to pathogenic populations. CSPG4 (chondroitin sulfate proteoglycan 4) is involved in cell migration and extracellular matrix remodeling [citations]—processes dysregulated in SLE-driven synovitis and organ infiltration. CSPG4 is already a clinical CAR-T target in melanoma and glioblastoma [citations], with Phase I/II trials demonstrating feasibility and manageable toxicity. ICAM1 (intercellular adhesion molecule 1) mediates lymphocyte-endothelial adhesion and is upregulated in chronic inflammation [citations]. ICAM1-targeted CAR-T has been tested in lymphoma [citation] and may be particularly suited to SLE given senescent cells' high adhesion molecule expression. CD38, while upregulated, was deprioritized due to expression on plasma cells and hematopoietic stem/progenitor cells, raising concerns for myelotoxicity and loss of protective immunity [Garelick 2025 discussion]. CD38 CAR-T in SLE warrants in vitro killing assays and safety profiling before clinical translation.

4.3: Senescence as a Biomarker for SLE Disease Activity and CAR-T Response

Beyond target identification, senescence scores may serve as a biomarker for SLE disease activity and CAR-T response prediction. Senescence score correlated strongly with SLEDAI ($p = 0.62$), matching or exceeding the predictive value of individual antibody or complement markers (anti-dsDNA $p \sim 0.4$, C3 $p \sim -0.45$) [citations]. In tissue samples, senescence score discriminated SLE from other inflammatory arthropathies (AUC 0.84), suggesting potential clinical utility as a diagnostic or stratification tool. The senescence biomarker captures multiple pathogenic mechanisms (cytokine production, adhesion molecule expression, metabolic dysregulation) in a single metric, potentially more predictive than single-molecule biomarkers [citations].

Speculatively, high senescence scores may identify SLE patients most likely to respond to CAR-T (high burden of targetable cells) and least likely to have treatment failure due to immune reconstitution (if senescence-driven dysfunction persists post-CAR-T). However, this remains hypothetical and requires prospective testing in CAR-T clinical trials.

4.4: Limitations & Future Directions

Several limitations warrant acknowledgment. First, scRNA-seq sample size is modest (10 SLE, 5 HC), potentially limiting generalizability, though bulk validation in >120 SLE samples provides reassurance. Second, clinical metadata in GEO datasets are incomplete; not all samples had SLEDAI scores, limiting disease activity correlation analysis (n=68/120 SLE). Third, this is purely in silico analysis; we have not performed CAR-T killing assays to functionally validate CSPG4 or ICAM1 targets in vitro. CAR-T cytotoxicity against senescent SLE immune cells is the critical next step before clinical translation. Fourth, tissue validation was limited to synovial biopsies; renal and CNS involvement (neuropsychiatric SLE) were not assessed, limiting conclusions about senescence in other lupus manifestations. Finally, the study is cross-sectional; longitudinal sampling during CAR-T therapy would clarify whether senescence predicts response and how immune reconstitution alters the senescence landscape. Future work should include functional CAR-T assays, prospective cohort studies in CAR-T-treated SLE patients, and expanded tissue profiling to test senescence as a pan-SLE therapeutic target.

4.5: Conclusion

In summary, we report that senescent immune cells accumulate in SLE and express distinct surface antigens including CSPG4 and ICAM1, which are upregulated relative to healthy controls and other inflammatory arthropathies. Senescence predicts disease activity and may identify CAR-T targets with improved specificity vs. pan-B cell depletion. Our findings support a senescence-guided approach to CAR-T target selection in SLE and establish the foundation for functional validation and clinical-translation studies. If confirmed, this work may establish senescence as a central therapeutic vulnerability in SLE, shifting treatment from non-selective immunosuppression toward targeted elimination of pathogenic, senescence-driven cellular populations.

5. ACKNOWLEDGEMENTS

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6. AUTHOR CONTRIBUTIONS

X.Y. designed the study, performed bioinformatic analyses, and drafted the manuscript. A.B. processed scRNA-seq data and contributed to figure preparation. C.D. performed statistical analyses. E.F. reviewed the manuscript and provided clinical interpretation. All authors approved the final version.

7. STATEMENTS AND DECLARATIONS

7.1: Ethical Considerations

Ethical approval: Not applicable. All data used in this study were from publicly available databases (GEO, NCBI) with prior ethical approval by original investigators. No new human subjects or tissues were enrolled.

7.2: Consent to Participate

Not applicable. This study used pre-existing, de-identified genomic data.

7.3: Consent for Publication

Not applicable.

7.4: Competing Interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

7.5: Funding

The author(s) received no specific funding for this research.

8. REFERENCES (~50 total, Sage Vancouver style)

Structure (by topic for clarity):

A. SLE Epidemiology & Pathogenesis (5–7 refs)

1. Rees et al. (worldwide incidence/prevalence)
2. Tsokos GC (SLE pathophysiology)
3. Others on B cell dysregulation, IFN-I

B. SLE Treatment & CAR-T (8–10 refs)

- Schett et al. (CD19 CAR-T in SLE, 2021)
- Garelick & Isenberg (CAR-T viewpoint, 2025)
- Rituximab, belimumab, anifrolumab trials
- CAR-T in hematologic malignancy (for comparison)

C. Senescence in Disease (5–7 refs)

- Saul et al. (SenMayo panel)
- Senescence in aging, autoimmunity
- SASP, cellular senescence mechanisms

D. Single-cell & Omics Methods (5–6 refs)

- Seurat, scRNA-seq integration
- AUCell, AddModuleScore
- Differential expression

E. Biomarkers in SLE (5–7 refs)

- Li et al. 2024 (IFITM3)
- Giani et al. 2021 (NP-SLE)
- Anti-dsDNA, C3/C4, SLEDAI

F. CAR-T Target Biology (5–6 refs)

- CSPG4 biology, CAR-T in melanoma
- ICAM1 in lymphocyte trafficking

- CD38, VCAM1, others

G. Additional Methods/Validation (4–5 refs)

- ROC analysis
- Statistical testing
- Others

Total: ~50 references